Western Governor's University

D213 Task 2: Sentiment Analysis

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As a data analyst, you will assess continuing data sources for their relevance to specific research questions throughout your career. The two organizations related to the given data set seek to analyze their operations and have collected variables of possible use to support decision-making processes.

Part I: Research Question

A. Describe the purpose of this data analysis by doing the following:

1. Summarize one research question that you will answer using neural network models and NLP techniques. Be sure the research question is relevant to a real-world organizational situation and sentiment analysis captured in your chosen data set(s).

Is it possible to build a neural network model trained on previous customer review data capable of processing customer review text records into a positive or negative sentiment classification prediction?

2. Define the objectives or goals of the data analysis. Be sure the objectives or goals are reasonable within the scope of the research question and are represented in the available data.

The goal of this data analysis will be to clean, tokenize, and analyze statistics on the provided vocabulary dataset to allow for modeling of a neural network capable of a binary classification sentiment analysis built using three datasets of customer reviews. This model will be capable of sentiment analysis on future datasets.

3. Identify a type of neural network capable of performing a text classification task that can be trained to produce useful predictions on text sequences on the selected data set.

The neural network will be built using a long-short term model as one layer. LSTM will be implemented for its ability to break down sequences and strength in retaining relevant information while effectively ‘forgetting’ the irrelevant.

(GeeksforGeeks, “What Is LSTM Long Short Term Memory?”)

Part II: Data Preparation

B. Summarize the data cleaning process by doing the following:

1. Perform exploratory data analysis on the chosen data set, and include an explanation of each of the following elements:

- presence of unusual characters (e.g., emojis, non-English characters)

There did not appear to be any special characters in the dictionary, but line 60 would replace any special characters and is implemented to be certain and ensure this code is applicable to future datasets.

- vocabulary size

The tokenizer.word_index function shows 5271 total words.

- proposed word embedding length

The proposed word embedding length will be 128.

- statistical justification for the chosen maximum sequence length

The maximum allowed sequence length is going to be set to the maximum string length of any record in the dataframe provided by the len().max() function.

2. Describe the goals of the tokenization process, including any code generated and packages that are used to normalize text during the tokenization process.

The goal of the tokenization process will be to convert the records containing cleaned text sequences into sequences of representative integers.

```
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
```

```
# Clean Data - replace new line with empty, lowercase row names, max length, repl spc
df[1] = df[1].map(lambda x : x.replace('\n',''))
df[0] = df[0].str.lower()
maxlen = df[0].str.len().max()
df[0] = df[0].str.replace('[^0-9a-zA-Z.,-/ ]', '')
words = {word for line in df[0]
    for word in line.rsplit(",",1)[0].split()}
words
vocablen = len(words)

df.rename(columns =  {0:'text',1:'sentiment_label'}, inplace = True)
text = df.text.values

#Tokenizer, convert text to sequences of integers
tokenizer = Tokenizer(num_words=vocablen)
tokenizer.fit_on_texts(text)
encoded_docs = tokenizer.texts_to_sequences(text)

# Pad sequences with 0s beginning of rows
padded_sequence = pad_sequences(encoded_docs, maxlen = maxlen+1)
print(padded_sequence)
pdd_sqnc_df = pd.DataFrame(padded_sequence)
pdd_sqnc_df.to_csv('C:/Users/velez/OneDrive/Desktop/School/D213/Task 2/padded.csv')

print(tokenizer.word_index)
```

3. Explain the padding process used to standardize the length of sequences. Include the following in your explanation:

- if the padding occurs before or after the text sequence

Zeroes are used to pad the text sequences by adding a record with the value 0. Each row will be extended by adding as many necessary records before the tokenized data until each row has as many records as the longest row.

- a screenshot of a single padded sequence

0	0	0	0	0	271	3	10	13	242	2	54	13	75	69

4. Identify how many categories of sentiment will be used and an activation function for the final dense layer of the network.

There will be two sentiment categories, positive and negative. The activation function for the dense layer is ‘sigmoid’.

5. Explain the steps used to prepare the data for analysis, including the size of the training, validation, and test set split (based on the industry average).

First, the three datasets, amazon, imdb and yelp are imported and have their datapoints split using ‘/’ to distinguish between records.

A function of ‘x_df[:-1]’ is run over each dataset to remove the final row. Records are then split on tabs.

The records are then added into a single newly defined dataframe, containing each record from each dataset.

The dataframe is then used to create a dictionary using pandas to_dict function.

(“Pandas.DataFrame.to_dict — Pandas 1.4.3 Documentation”)

The counter function is implemented to list and index each vocabulary word in the dataframe.

(GeeksforGeeks, “Find Frequency of Characters in Python”)

All indicators of new line (\n) are removed. The first column (column 0) is adjusted to be entirely lowercase. The maximum length for a record in the dataframe is set equal to the observed maximum string length in column 0.

A line is implemented to remove all characters that are not the defined a-z and 0-9- essentially removing all special characters and leaving that space empty.

Then the code will break each sentence into individual words and keep them in a set defined as ‘words’.

The numeric columns are renamed to ‘text’ and ‘sentiment_label’. Text values are extracted from the dataframe.

The tokenizer is initiated and implemented on the previously defined text values, converting the text into integers indexed to the vocabulary word they represent. This is done in sequence, so each record will be a ‘sequence’ of integers representing a sentence.

(“Tf.keras.preprocessing.text.Tokenizer | TensorFlow Core V2.3.0”)

The dataset is now padded, meaning all rows will have an equal amount of records (columns) by setting any records exceeding the observed record amount per row to 0, up to the value of the longest row plus one. The padding is added to the beginning of each row.

Finally, the dataset can be split 90/10 for a train/test split. X train will have a shape of (2700,480), and X test a shape of (300,480). Y test has a shape of (2700,) and Y train a shape of (300,). The validation data will be set to equal (x_test, y_train) which is ((2700,480),(300,)).

6. Provide a copy of the prepared data set.

See file attached: “ReadyForAnalysis.csv”

Part III: Network Architecture

C. Describe the type of network used by doing the following:

1. Provide the output of the model summary of the function from TensorFlow.

```
...: print(model.summary())
Model: "sequential"
```

Layer (type)	Output Shape	Param #
embedding (Embedding)	(None, 480, 120)	748,800
lstm (LSTM)	(None, 60)	45,360
dense (Dense)	(None, 1)	61

```
Total params: 794,221 (3.03 MB)
Trainable params: 794,221 (3.03 MB)
Non-trainable params: 0 (0.00 B)
None
```

(Mayo)

2. Discuss the number of layers, the type of layers, and the total number of parameters.

This sequential model from tensorflow includes three layers: Embedding with 748,800 parameters, LSTM with 45,360, and Dense with 61. There are 794,221 total parameters, all of which are trainable. One goal of this step was to ensure there were no remaining non-trainable parameters, and this was achieved.

3. Justify the choice of hyperparameters, including the following elements:

- activation functions

The LSTM layer will have an activation function of ‘Tanh’ to run on non-linear, higher complexity functions. The Dense Layer will have an activation function of sigmoid as it is being used for a binary classification model.

(GeeksforGeeks, “Tanh Activation in Neural Network”)

- number of nodes per layer

Each layer will have two nodes in addition to the Layer type classification.

(GeeksforGeeks, “Neural Network Node”)

- loss function

Model.compile() will have a loss function of binary cross-entropy, as this sentiment analysis will have a binary classification.

(“Binary Cross Entropy/Log Loss for Binary Classification”)

(“Keras Documentation: Model Training APIs”)

- optimizer

Adam is chosen as the optimizer as it will best allow for a variable learning rate in the neural network.

- stopping criteria

The stopping criteria will allow two epochs with no improvement before it stops training.

(“Keras Documentation: EarlyStopping”)

- evaluation metric

The selected metric is accuracy in the model.compile() function. Validation accuracy and training accuracy will be modeled against one another on a line graph, along with a second line graph corresponding to the models’ losses. There is also a confusion matrix plotting predicted sentiment labels against true sentiment labels.

Part IV: Model Evaluation

D. Evaluate the model training process and its relevant outcomes by doing the following:

1. Discuss the impact of using stopping criteria to include defining the number of epochs, including a screenshot showing the final training epoch.

The model.fit function has the maximum number of epochs set to four, with the caveat that if two epochs consecutively show no improvement, no more epochs will be run.

```
85/85 ━━━━━━━━━━━━━━━━━━━━ 11s 135ms/step - accuracy: 0.9981 - loss: 0.0129 -
val_accuracy: 0.8300 - val_loss: 0.7846
Traceback (most recent call last):
```

2. Assess the fitness of the model and any actions taken to address overfitting.

The final model experiences a training accuracy loss decrease with a validation loss increase, indicative of an overfitting model. This could indicate the model has memorized the training data but is not fully capable of making predictions on other datasets as accurately.

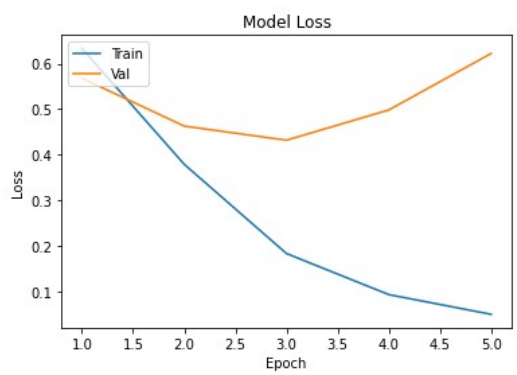
Initially, the model was run using five epochs, with the fifth epoch resulting in validation accuracy decrease. This is addressed by setting the fourth epoch as the final, as this fourth model boasts the highest validation and training accuracy.

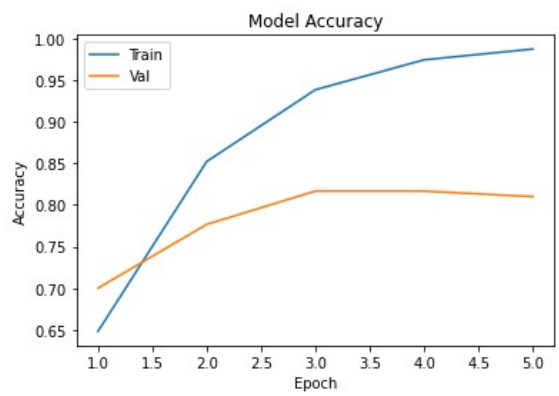
```
In [10]:
    ...: early_stopping_monitor = EarlyStopping(patience=2)
    ...: history = model.fit(x_train,y_train, epochs=5, batch_size=32,
validation_data=(x_test,y_test))
Epoch 1/5
85/85 ━━━━━━━━━━━━━━━━━━━━ 15s 143ms/step - accuracy: 0.6481 - loss: 0.6332 -
val_accuracy: 0.7000 - val_loss: 0.5680
Epoch 2/5
85/85 ━━━━━━━━━━━━━━━━━━━━ 12s 141ms/step - accuracy: 0.8522 - loss: 0.3796 -
val_accuracy: 0.7767 - val_loss: 0.4631
Epoch 3/5
85/85 ━━━━━━━━━━━━━━━━━━━━ 12s 145ms/step - accuracy: 0.9385 - loss: 0.1842 -
val_accuracy: 0.8167 - val_loss: 0.4322
Epoch 4/5
85/85 ━━━━━━━━━━━━━━━━━━━━ 20s 239ms/step - accuracy: 0.9744 - loss: 0.0943 -
val_accuracy: 0.8167 - val_loss: 0.4985
Epoch 5/5
85/85 ━━━━━━━━━━━━━━━━━━━━ 22s 253ms/step - accuracy: 0.9874 - loss: 0.0512 -
val_accuracy: 0.8100 - val_loss: 0.6220
```

```
    ....
    ...: model.evaluate(x_test, y_test)
Epoch 1/4
85/85 ━━━━━━━━━━━━━━━━━━━━ 13s 148ms/step - accuracy: 0.9889 - loss: 0.0348 -
val_accuracy: 0.8100 - val_loss: 0.6648
Epoch 2/4
85/85 ━━━━━━━━━━━━━━━━━━━━ 12s 136ms/step - accuracy: 0.9922 - loss: 0.0305 -
val_accuracy: 0.8200 - val_loss: 0.7164
Epoch 3/4
85/85 ━━━━━━━━━━━━━━━━━━━━ 12s 138ms/step - accuracy: 0.9944 - loss: 0.0215 -
val_accuracy: 0.8167 - val_loss: 0.7592
Epoch 4/4
85/85 ━━━━━━━━━━━━━━━━━━━━ 11s 135ms/step - accuracy: 0.9981 - loss: 0.0129 -
val_accuracy: 0.8300 - val_loss: 0.7846
Traceback (most recent call last):
```

(“Training and Validation Loss in Deep Learning”)

3. Provide visualizations of the model’s training process, including a line graph of the loss and chosen evaluation metric.





4. Discuss the predictive accuracy of the trained network using the chosen evaluation metric from part D3.

The training accuracy of the model rises to 99.8% on the final epoch, with a validation accuracy of 83%. This indicates the model is predicting the data it is trained on nearly perfectly, with a significant drop off but still a strong prediction when applied on non-training data. Each epoch consistently decreases the loss of the training data predictions, but loss on the validation data remains mostly consistent.

Part V: Summary and Recommendations

E. Provide the code you used to save the trained network within the neural network.

```
model.save("Trained_Model.keras")
```

F. Discuss the functionality of your neural network, including the impact of the network architecture.

This neural network is a sentiment analysis model, which takes datasets of customer review text records and converts to counts of positive or negative sentiment prediction.

This model's input layer is embedding, which will train the model on the integers representing the words converted from the tokenizer process. The embedding layer has a trade off in attempting to balance capturing enough dimensions without overfitting by attempting to capture too high a number.

The middle layer, LSTM, will train the model based on relationships between sequences of words. LSTM determines which information the model should forget or output into the next level of processing.

The final Dense layer of the model will use the sigmoid activation function to output a binary classification prediction of that record showing positive or negative sentiment. This is also called the fully connected layer because each neuron in this final layer is a direct result of the previous layers passing on and processing information

(GeeksforGeeks, "What Are Embeddings in Machine Learning?")

(GeeksforGeeks, "What Is LSTM Long Short Term Memory?")

(GeeksforGeeks, "What Is Fully Connected Layer in Deep Learning?")

G. Recommend a course of action based on your results.

Companies may not be comfortable with 83% accuracy on the validation data. The model's validation accuracy is likely improvable by testing dropout layers and additional hyperparameter tuning, which was implemented, but all tests resulted in less accurate models. When the company is pleased with the validation

accuracy, it would be recommended to implement this model on future datasets containing similar review records to quickly provide useful statistics on positive and negative sentiments.

Part VI: Reporting

H. Show your neural network in an industry-relevant interactive development environment (e.g., a Jupyter Notebook). Include a PDF or HTML document of your executed notebook presentation.

See files attached: ‘BrandonVelezD213Task2.pdf’, ‘BrandonVelezD213Task2.py’, ‘BrandonVelezD213Task2.ipynb’

I & J : **Works Cited**

GeeksforGeeks. “Binary Cross Entropy/Log Loss for Binary Classification.” *GeeksforGeeks*, 27 May 2024, www.geeksforgeeks.org/deep-learning/binary-cross-entropy-log-loss-for-binary-classification/.

---. “Find Frequency of Characters in Python.” *GeeksforGeeks*, 22 Nov. 2018, www.geeksforgeeks.org/python/python-frequency-of-each-character-in-string/. Accessed 27 Oct. 2025.

---. “Neural Network Node.” *GeeksforGeeks*, 13 Sept. 2024, www.geeksforgeeks.org/deep-learning/neural-network-node/#. Accessed 4 Nov. 2025.

---. “Tanh Activation in Neural Network.” *GeeksforGeeks*, 14 Feb. 2025, www.geeksforgeeks.org/deep-learning/tanh-activation-in-neural-network/. Accessed 10 Nov. 2025.

---. “Training and Validation Loss in Deep Learning.” *GeeksforGeeks*, 18 Sept. 2024, www.geeksforgeeks.org/deep-learning/training-and-validation-loss-in-deep-learning/#. Accessed 6 Nov. 2025.

---. “What Are Embedding in Machine Learning?” *GeeksforGeeks*, 15 Feb. 2024, www.geeksforgeeks.org/machine-learning/what-are-embeddings-in-machine-learning/.

---. “What Is Fully Connected Layer in Deep Learning?” *GeeksforGeeks*, 27 May 2024, www.geeksforgeeks.org/deep-learning/what-is-fully-connected-layer-in-deep-learning/.

---. “What Is LSTM Long Short Term Memory?” *GeeksforGeeks*, 16 Jan. 2019, www.geeksforgeeks.org/deep-learning/deep-learning-introduction-to-long-short-term-memory/.

“Keras Documentation: EarlyStopping.” *Keras.io*, keras.io/api/callbacks/early_stopping/.

“Keras Documentation: Model Training APIs.” *Keras.io*, keras.io/api/models/model_training_apis/.

“Pandas.DataFrame.to_dict — Pandas 1.4.3 Documentation.” *Pandas.pydata.org*, pandas.pydata.org/docs/reference/api/pandas.DataFrame.to_dict.html.

“Tf.keras.preprocessing.text.Tokenizer | TensorFlow Core V2.3.0.” *TensorFlow*, www.tensorflow.org/api_docs/python/tf/keras/preprocessing/text/Tokenizer.